

The 007786 scans for valid object data:

- Object (sprite) code,
- palette,
- position and
- attributes

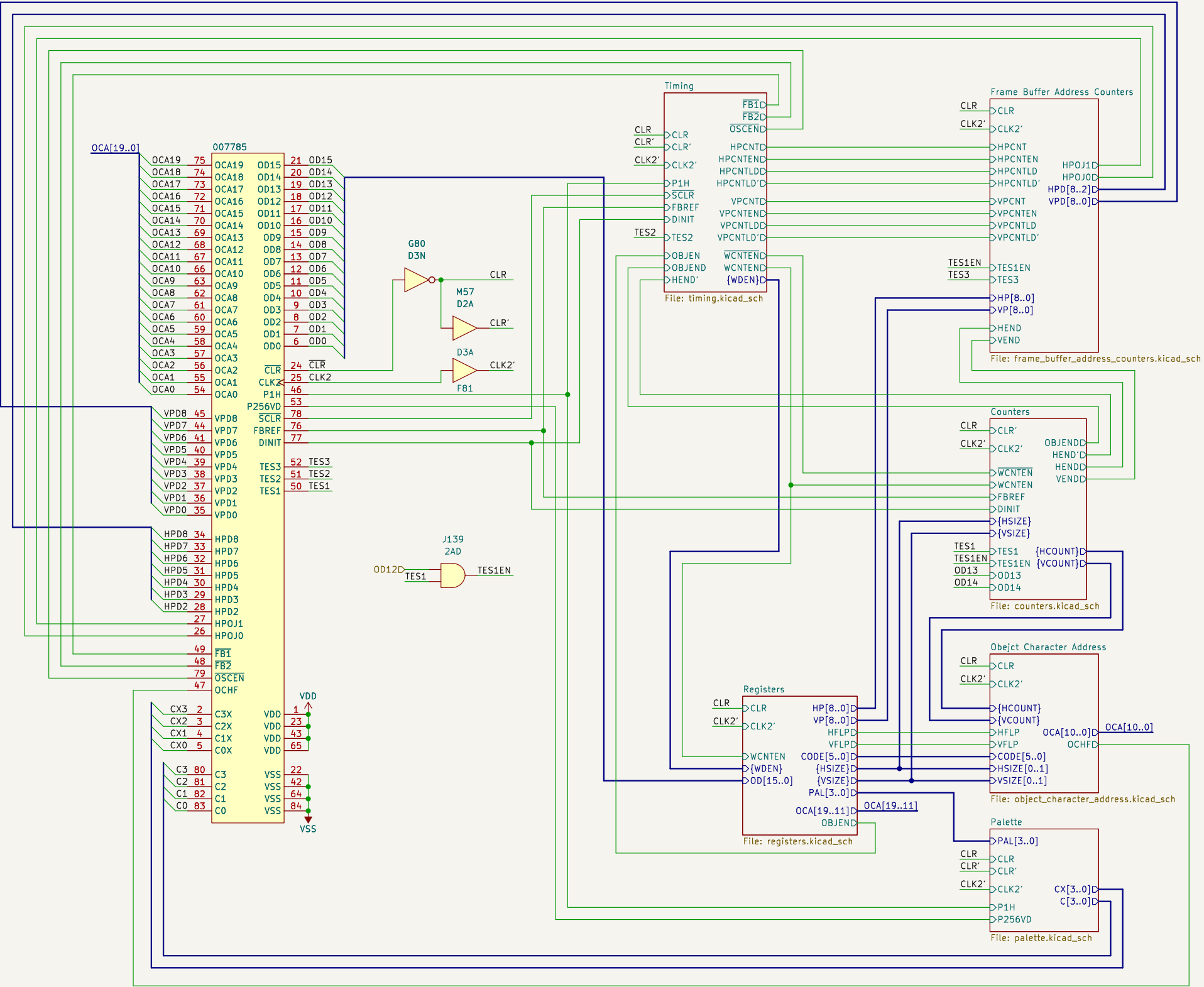
It translates it into:

- Individual object character (tiles) color code,
- Palette values
- object character address

This is copied into the frame buffers with the 007786.

1. SCLR (OSCANCLR) starts the obejct scanning process.
2. OSCEN is activated when each of the four words to be scanned are to be read. The 007783 increments the address.

The 007786 uses an OKI 79V000 gate cell array with 3289 unit cells.



Ulf Skutnabba, twitter: @skutis77

Sheet: /  
File: 007785.kicad\_sch

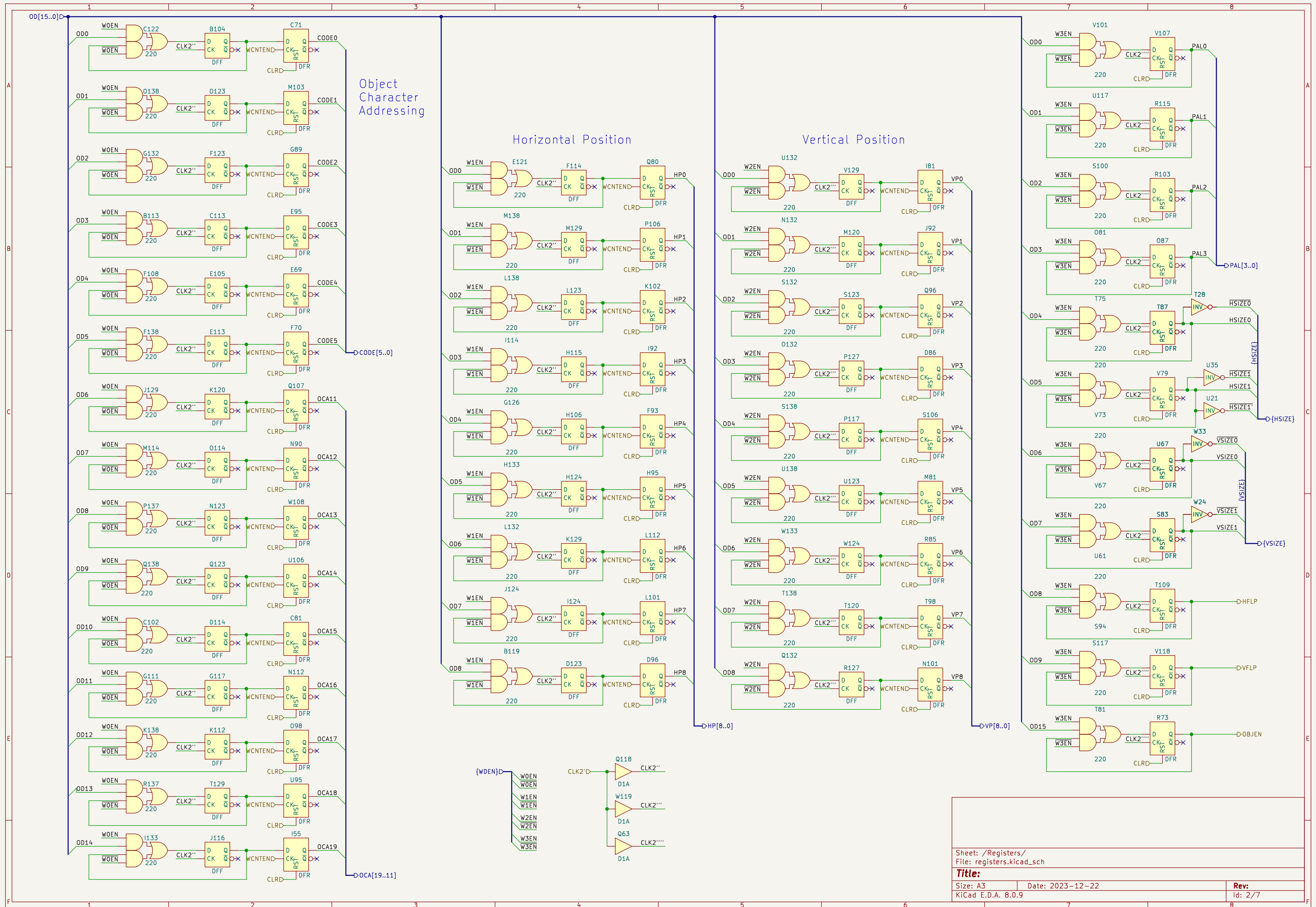
Title: Konami 007785

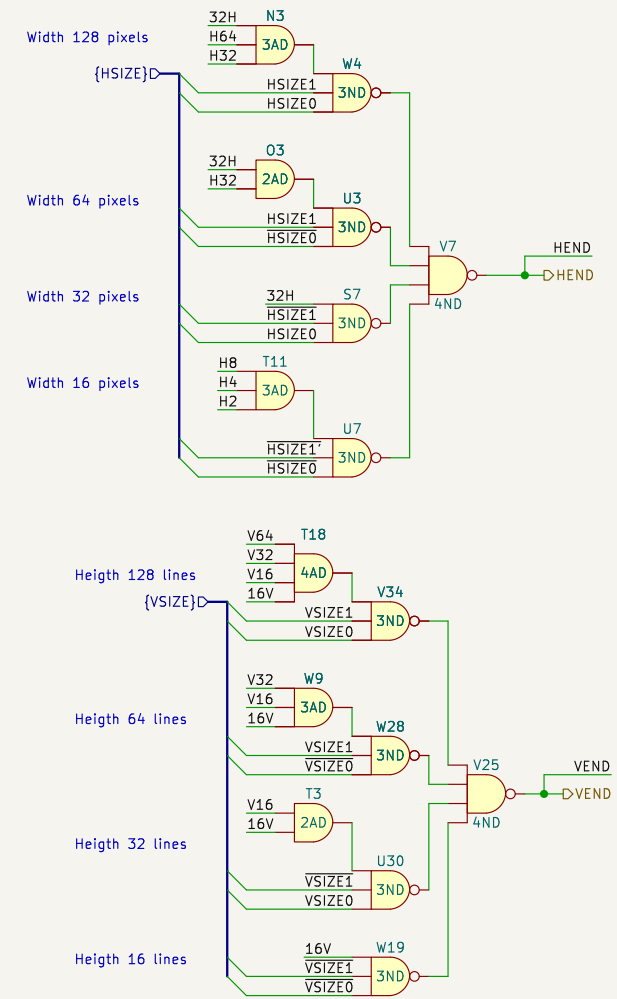
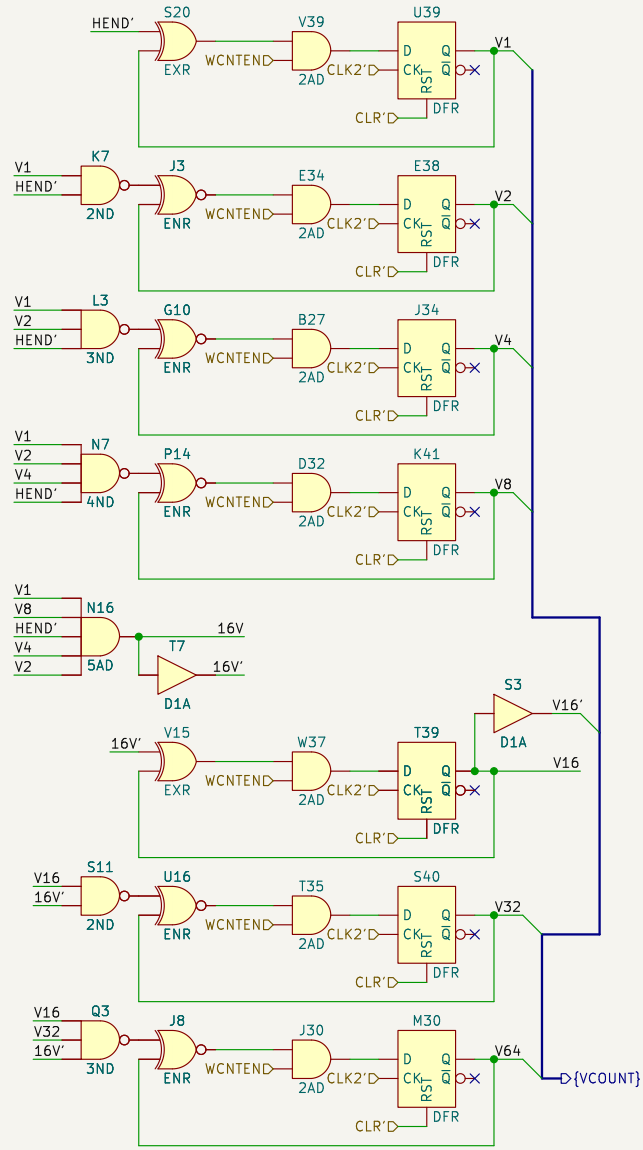
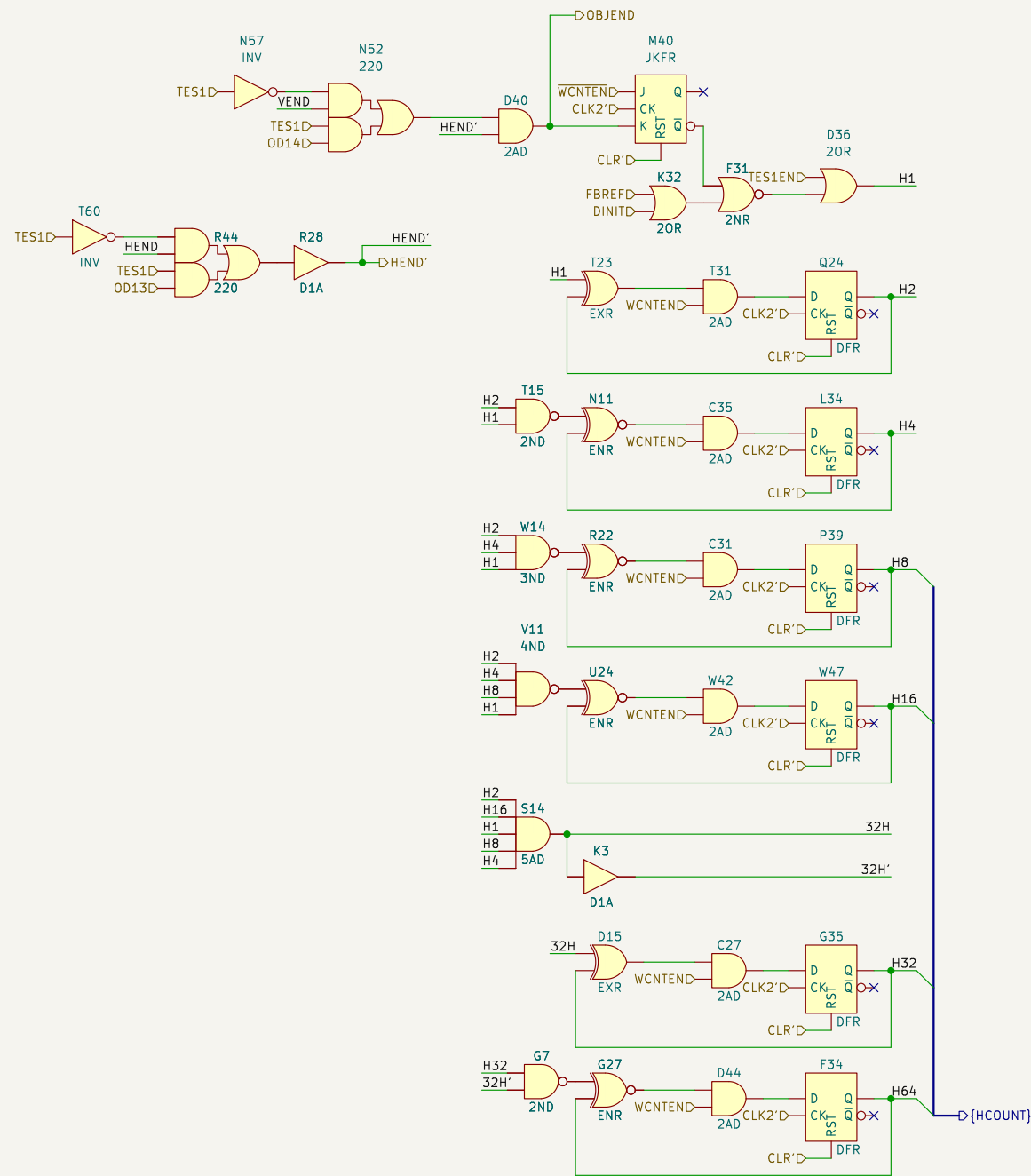
Size: A3 Date: 2023-12-22

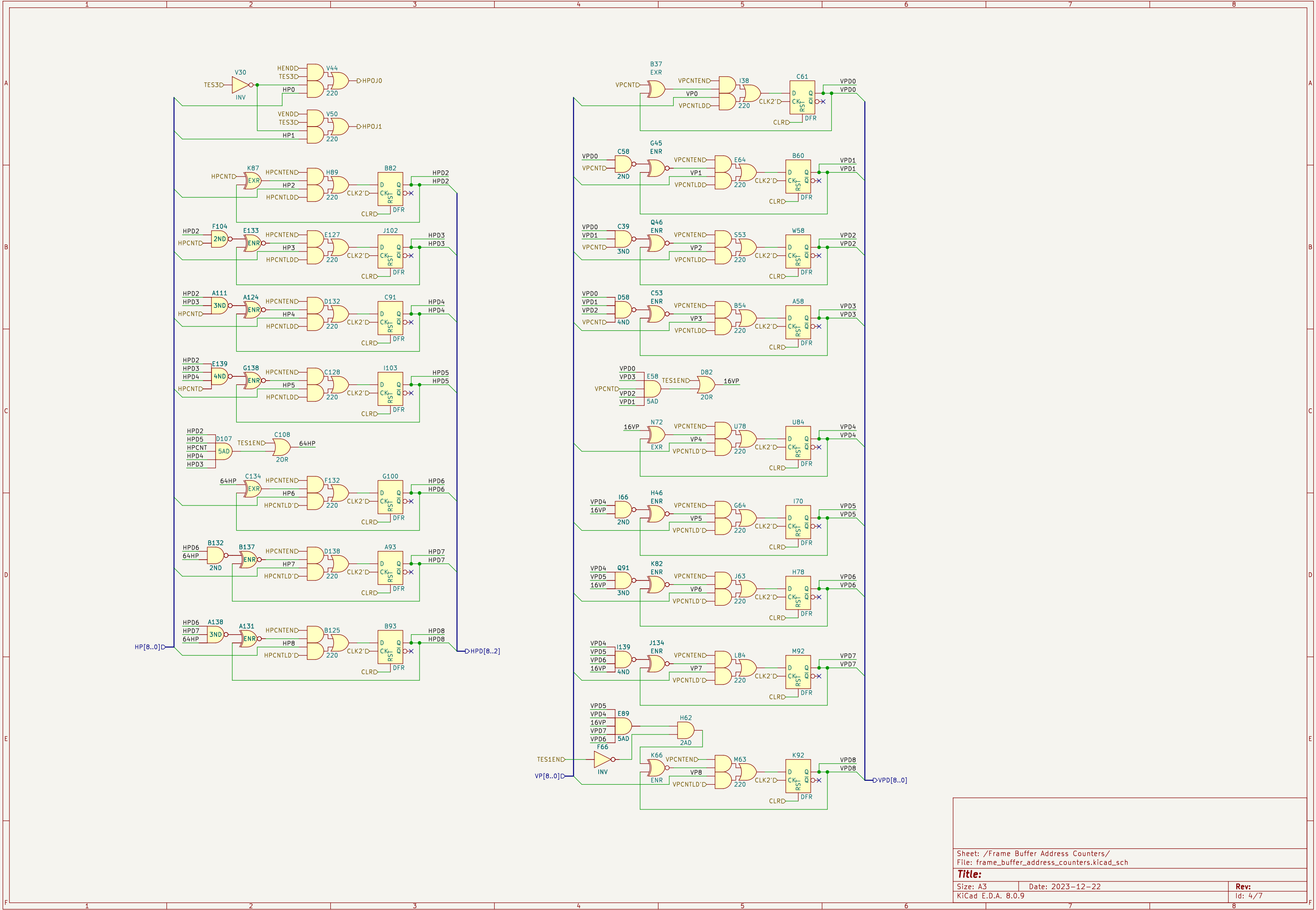
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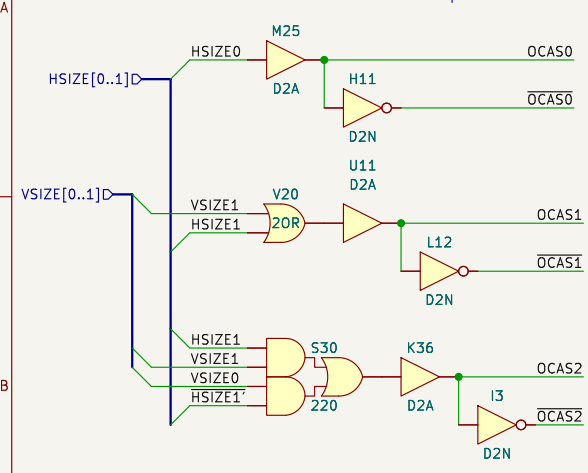




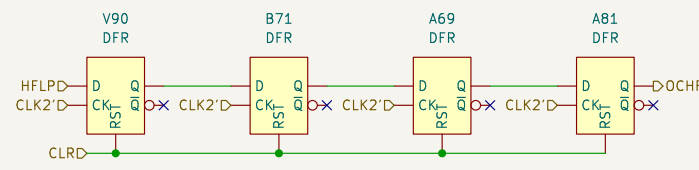
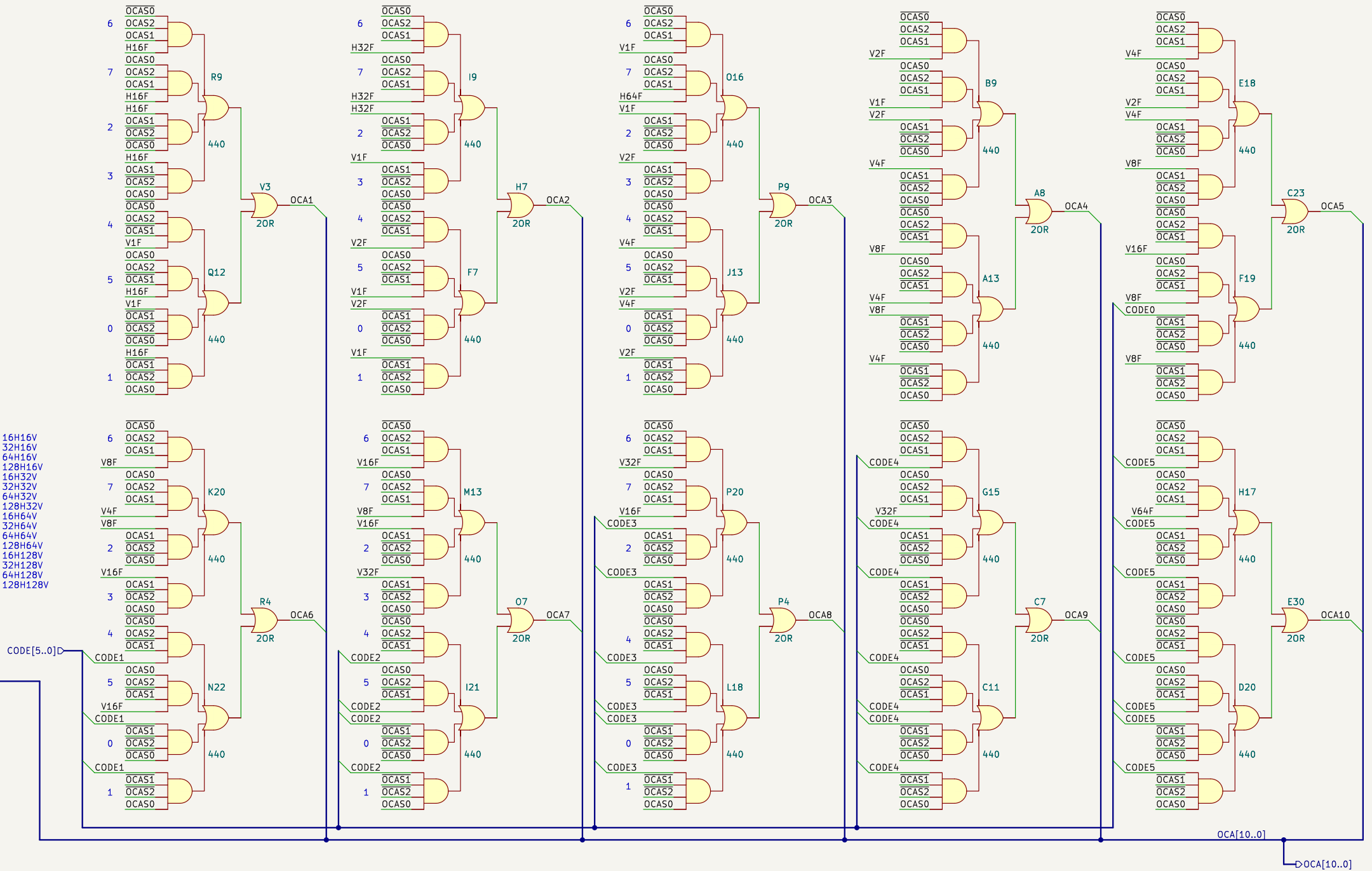
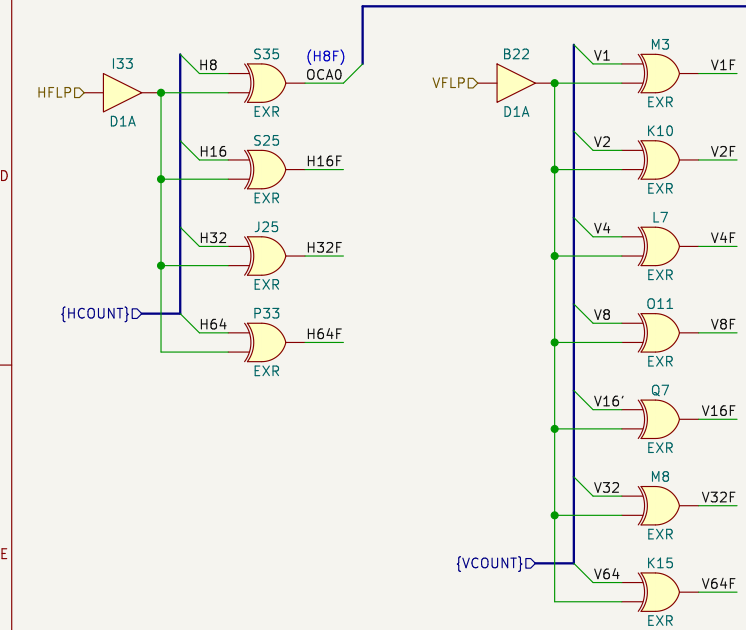


Object Character Select Address

Object (Sprite) tiles are called characters on the twin16 platform.



| VSIZE1 | VSIZE0 | VSIZE | HSIZE1 | HSIZE0 | HSIZE | OCA2 | OCA1 | OCA0 |   |          |
|--------|--------|-------|--------|--------|-------|------|------|------|---|----------|
| 0      | 0      | 16    | 0      | 0      | 16    | 0    | 0    | 0    | 0 | 16H16V   |
| 0      | 0      | 16    | 1      | 0      | 32    | 0    | 0    | 1    | 1 | 32H16V   |
| 0      | 0      | 16    | 1      | 1      | 64    | 0    | 1    | 0    | 0 | 64H16V   |
| 0      | 0      | 16    | 1      | 1      | 128   | 0    | 1    | 1    | 1 | 128H16V  |
| 0      | 1      | 32    | 0      | 0      | 16    | 1    | 0    | 0    | 0 | 16H32V   |
| 0      | 1      | 32    | 0      | 1      | 32    | 1    | 0    | 1    | 0 | 32H32V   |
| 0      | 1      | 32    | 0      | 1      | 64    | 1    | 1    | 0    | 1 | 64H32V   |
| 0      | 1      | 32    | 0      | 1      | 128   | 0    | 1    | 1    | 1 | 128H32V  |
| 1      | 0      | 64    | 0      | 0      | 16    | 0    | 1    | 0    | 0 | 16H64V   |
| 1      | 0      | 64    | 0      | 1      | 32    | 0    | 1    | 1    | 1 | 32H64V   |
| 1      | 0      | 64    | 1      | 0      | 64    | 1    | 1    | 0    | 0 | 64H64V   |
| 1      | 0      | 64    | 1      | 1      | 128   | 1    | 1    | 1    | 1 | 128H64V  |
| 1      | 1      | 128   | 0      | 0      | 16    | 1    | 1    | 0    | 0 | 16H128V  |
| 1      | 1      | 128   | 0      | 1      | 32    | 1    | 1    | 1    | 1 | 32H128V  |
| 1      | 1      | 128   | 1      | 0      | 64    | 1    | 1    | 0    | 0 | 64H128V  |
| 1      | 1      | 128   | 1      | 1      | 128   | 1    | 1    | 1    | 1 | 128H128V |



| Size    |       |
|---------|-------|
| 16H16V  | H8F   |
| 32H16V  | H16F  |
| 64H16V  | H32F  |
| 128H16V | H64F  |
| 16H32V  | H16F  |
| 32H32V  | H32F  |
| 64H32V  | H64F  |
| 128H32V | H128F |
| 16H64V  | H16F  |
| 32H64V  | H32F  |
| 64H64V  | H64F  |
| 128H64V | H128F |

| OCA0  | OCA1 | OCA2 | OCA3 | OCA4 | OCA5  | OCA6  | OCA7  | OCA8  | OCA9  | OCA10  |
|-------|------|------|------|------|-------|-------|-------|-------|-------|--------|
| H8F   | V1F  | V2F  | V4F  | V8F  | CODE0 | CODE1 | CODE2 | CODE3 | CODE4 | CODE5  |
| H16F  | V1F  | V2F  | V4F  | V8F  | CODE1 | CODE2 | CODE3 | CODE4 | CODE5 | CODE6  |
| H32F  | V1F  | V2F  | V4F  | V8F  | CODE2 | CODE3 | CODE4 | CODE5 | CODE6 | CODE7  |
| H64F  | V1F  | V2F  | V4F  | V8F  | CODE3 | CODE4 | CODE5 | CODE6 | CODE7 | CODE8  |
| H128F | V1F  | V2F  | V4F  | V8F  | CODE4 | CODE5 | CODE6 | CODE7 | CODE8 | CODE9  |
| H256F | V1F  | V2F  | V4F  | V8F  | CODE5 | CODE6 | CODE7 | CODE8 | CODE9 | CODE10 |

Sheet: /Obejct Character Address/  
File: object\_character\_address.kicad\_sch

| Title:             |                  |         |
|--------------------|------------------|---------|
| Size: A3           | Date: 2023-12-22 | Rev:    |
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